

Chapter-2 Network Models

1. Define Network

A network is a combination of hardware and software that sends data from one location to another.

2. Layered Tasks Concept

- Communication is divided into layers for simplicity.
- Example: Sending a letter involves:
 - **Higher Layer:** Writing and addressing the letter
 - **Middle Layer:** Carrier picks and delivers to post office
 - **Lower Layer:** Sorting and transporting to destination

3. OSI Model Overview

- Developed by ISO (International Standards Organization)
- Not a protocol — it's a framework for designing flexible, interoperable networks
- Purpose: Enable communication between different systems without changing hardware/software logic

4. 7 Layers of OSI Model

Layer No.	Name	Function Summary
1	Physical	Transmits raw bits over medium
2	Data Link	Reliable node-to-node delivery
3	Network	Source-to-destination delivery
4	Transport	Process-to-process delivery
5	Session	Manages sessions/dialogs
6	Presentation	Data format, encryption, compression
7	Application	User access to network services

5. Physical Layer Responsibilities

- Bit representation (electrical/optical signals)
- Data rate and synchronization
- Line configuration (point-to-point/multipoint)
- Topology (mesh, star, ring, bus)
- Transmission mode (simplex, half/full duplex)

6. Data Link Layer Responsibilities

- Framing and physical addressing
- Flow and error control
- Access control for shared links

7. Network Layer Responsibilities

- Logical addressing (IP)
- Routing across multiple networks

8. Transport Layer Responsibilities

- Port addressing (service-point)
- Segmentation and reassembly
- Connection control (connectionless/oriented)
- End-to-end flow and error control

9. Session Layer Responsibilities

- Dialog control (half/full duplex)
- Synchronization (checkpoints for recovery)

10. Presentation Layer Responsibilities

- Translation between encoding formats
- Encryption/decryption
- Compression (especially for multimedia)

11. Application Layer Services

- Network virtual terminal
- File transfer and management
- Email and directory services

12. Protocols by Layer (Examples)

Layer	Protocols (Examples)
Physical	RS232, USB, DSL
Data Link	PPP, ARP, HDLC
Network	IP, ICMP, OSPF
Transport	TCP, UDP, SCTP
Session	RPC, NetBIOS, ZIP
Presentation	SSL, TLS, MPEG
Application	HTTP, FTP, SMTP

13. OSI vs TCP/IP Model

Feature	OSI Model	TCP/IP Model
Layer Separation	Clear (7 layers)	Mixed (4 layers)
Bottom Layers	Physical + Data Link	Combined as Link Layer
Transport Layer	Connection-oriented only	Both connection types
Header Size	Min 5 bytes	Min 20 bytes